

Element A

Context:

An engineering teacher decides to take his family on a vacation. One of the three children is young and has a difficult time locking public restroom doors. It is also known that the child has trouble using the restroom when other people are watching. Without an effective way to keep the door closed, it would be very difficult for the primary stakeholders to take this trip with a child that will need to use the restroom approximately 4-7 times a day¹.

Problem Statement:

Our team must design a device that will be used on public restroom doors that will effectively keep the door shut while the child is in the stall without the child being part of the solution.

Problem Elaboration:

Currently, the child is not able to utilize public restrooms because of his inability to use the locks in the restrooms, or because the restroom door does not have a lock included on the door. The primary stakeholder made it clear that the child will not be a part of the solution, and the lock will have to be accessed outside of the door by the parent.

Our primary stakeholder is Mr.Kiser and his child. In general, parents and guardians who have responsibility for their child's safety and well being while using a public restroom.

The secondary stakeholder is the parent not utilizing the lock, siblings of the child, other family members who could be delayed or have their trip affected by a bathroom delay, and staff members of the bathroom facilities. Staff members benefit from having their facilities clean² and tidy while other members of the public waiting to use the restroom can also be considered secondary stakeholders.

In an interview with one of our primary stakeholders, Parent A of the child, we asked the following questions. Note that the responses are paraphrased and not directly quoted.

1. What materials will we use?
 - We will use materials provided by the class.
2. How long will it hold the door closed?
 - The door will be able to be held closed for 3-4 minutes.
3. How light will the lock be?
 - 2-3 pounds.
4. Will there be any dangerous materials or will the lock itself be dangerous?
 - There will be no sharp metal nor anything that could injure the user.
5. How much room will it take to hold the lock?
 - It would be able to fit in a standard pocket size.
6. Can the child operate the lock with a disability such as being one handed?
 - Yes the child will be able to operate the lock with one hand.
7. How long will it take to apply the lock?
 - Ideally the lock would take 5 seconds to apply to the stall.
8. How long will the construction take?
 - Around 7 business days.
9. How many times will the lock be used throughout the day?
 - The lock will need to be used 5 to 6 times a day.

The stakeholder also noted that they had a wrist injury on their left arm, which prohibits us from using designs that need two hands to apply.

Measurements

We measured both of the boys bathroom stalls; measurements are below.

STALL 1

- Front Door Length - 147.5 cm
- Front Door Width - 59.6 cm
- Front of Stall - 208.5 cm
- Side of Stall - 155.5 cm
- Lock Key Width - 8 cm
- Lock Key Height - 11 cm
- Lock Key Hole Width - 7 cm
- Lock Key Hole Height - 6.5 cm

STALL 2 (Handicaped)

- Front Door Length - 147.5 cm
- Front Door Width - 90 cm
- Front of Stall - 208.5 cm
- Side of Stall - 155.5 cm
- Lock Key Width - 7 cm
- Lock Key Height - 6.5 cm
- Lock Key Hole Width - 5 cm
- Lock Key Hole Height - 6.5 cm

Problem Justification:

Globally, there are plenty of demographics that have trouble accessing locks on restroom doors, including persons who have no arms, no hands, or difficulty utilizing many standard locks on restrooms including rotating and or the frictional force of the lock being unlocked. Most public bathrooms are not designed for children usage either, as some locks can be anywhere from This now makes the problem much larger than our specific situation.

Citations:

1. Children's Dayton 06-30-2025, "When to Be Concerned", Children's Dayton, <https://childrensdayton.org/resources/when-to-be-concerned/frequent-urination-in-kids/>
2. Enviro-Master 08-20-2025, "Why Clean Public Restrooms Important", Enviro-Master, <https://enviro-master.com/commercial-cleaning-information/why-are-clean-restrooms-important/>